

Dittert's conjecture in dimension 10

An exact computer-assisted boundary-exclusion proof

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Status. This is an exact, reproducible working proof. It has not yet undergone independent peer review. The finite dimension-specific calculations are checked by two separately implemented standard-library verifiers.

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Abstract

For a nonnegative $n \times n$ matrix A whose entries sum to n , let r_i, c_j be its row and column sums and put

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \text{per } A.$$

We prove Dittert's conjecture in dimension $n = 10$. As in the proofs for $n = 11, 12, 13$, a maximal Li scaling reduces the boundary problem to marginal and proper tight cuts. Proper cuts are excluded by exact staircase-face bounds. For a marginal cut we add the stationarity equation at its minimum row or column. It forces a line-sum deviation on the opposite side, allowing the shared row-column product deficit to exclude the last case. The two univariate inequalities are certified exactly, first by rational Sturm sequences and independently by exact Ryser interpolation and Bernstein positivity.

1 Statement and published inputs

Let

$$K_n = \left\{ A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j} a_{ij} = n \right\}, \quad \gamma_n = \frac{n!}{n^n}.$$

For $A \in K_n$, write

$$R = \prod_{i=1}^n r_i, \quad C = \prod_{j=1}^n c_j, \quad P = \text{per } A.$$

Thus $\Phi(A) = R + C - P$ and

$$\Phi(J_n/n) = 2 - \gamma_n.$$

Theorem 1.1. *For every $A \in K_{10}$,*

$$\Phi(A) \leq 2 - \frac{10!}{10^{10}},$$

with equality if and only if $A = J_{10}/10$.

We use the following published results.

1. Hwang proved that the only possible positive Φ -maximizer in K_n is J_n/n [5].
2. Cheon and Wanless proved that a partly decomposable matrix cannot maximize Φ , and that a maximizer cannot have its complete zero set equal, up to row and column permutations, to one proper rectangular block [1, Theorems 3.2 and 3.7].
3. Li's cut characterization says that a nonnegative $n \times n$ matrix S is doubly superstochastic if and only if

$$\sum_{i \in I, j \in J} s_{ij} \geq |I| + |J| - n$$

for all $I, J \subseteq [n]$ [6]; see also [1, Lemma 2.2].

4. On a staircase face of the Birkhoff polytope, the barycenter minimizes the permanent [4].
5. Minc's averaging theorem for permanent minimizers with repeated support rows and columns gives the block form used below [7, Theorem 1]; see also [9, Section 2].
6. The van der Waerden permanent theorem gives $\text{per } B \geq \gamma_n$ for every doubly stochastic B , with equality only at J_n/n [2, 3].

The stationarity calculation that is new to the marginal-cut argument is proved directly in Section 5; it is not an additional external assumption.

2 Boundary structure and the shared deficit

Because K_n is compact, Φ has a global maximizer. By Hwang's positive-support theorem, a nonuniform maximizer must lie on the boundary.

Lemma 2.1 (Two independent zeroes). *Every boundary Φ -maximizer has two zero entries in distinct rows and distinct columns.*

Proof. Suppose a nonempty zero set contains no such pair. Fix a zero at (i, j) . Every other zero shares row i or column j . If there were both a zero (i, j') , $j' \neq j$, and a zero (i', j) , $i' \neq i$, those two zeroes would be in distinct rows and columns. Hence all zeroes lie in one row or all lie in one column.

If that row or column is entirely zero, the matrix is partly decomposable. Otherwise its complete zero set is one proper rectangular block. Both cases are excluded for a maximizer by Cheon–Wanless. $\square$

Assume for contradiction that a boundary maximizer A satisfies

$$\Phi(A) \geq 2 - \gamma_n.$$

The row sums and column sums each have arithmetic mean 1, so AM–GM gives $R, C \leq 1$. Consequently $P \leq \gamma_n$. Write

$$P = \gamma_n - \delta, \quad 0 \leq \delta \leq \gamma_n,$$

and put $R = 1 - \rho$, $C = 1 - \sigma$. The assumed lower bound on Φ gives

$$\rho + \sigma \leq \delta. \tag{1}$$

In particular,

$$R \geq 1 - \delta, \quad C \geq 1 - \delta, \quad RC = (1 - \rho)(1 - \sigma) \geq 1 - \delta. \tag{2}$$

3 The maximal Li scaling

For $I, J \subseteq [n]$, let $s(A[I, J])$ be the sum of the indicated submatrix, and define

$$\lambda = \min_{\substack{I, J \subseteq [n] \\ k := |I| + |J| - n > 0}} \frac{s(A[I, J])}{k}. \quad (3)$$

A maximizer is fully indecomposable by the Cheon–Wanless exclusion. Hence every numerator in (3) is positive, so $\lambda > 0$. Taking $I = J = [n]$ shows $\lambda \leq 1$.

Li's theorem says that A/λ is doubly superstochastic. Thus there is a doubly stochastic matrix B such that

$$\lambda B \leq A \quad \text{entrywise}, \quad P = \text{per } A \geq \lambda^n \text{per } B. \quad (4)$$

Choose I, J attaining (3), and set

$$u = n - |I|, \quad v = n - |J|, \quad k = n - u - v > 0.$$

There are three cases: the whole cut $(u, v) = (0, 0)$, a marginal cut where exactly one of u, v is zero, and a proper cut where $u, v \geq 1$.

If $u = v = 0$, then $\lambda = 1$. Since $B \leq A$ and both matrices have total sum n , $A = B$ is doubly stochastic. The van der Waerden theorem and $P \leq \gamma_n$ then force $A = J_n/n$, contrary to the boundary assumption.

4 The two-independent-zero permanent gap

Let $\Omega_n^{(2)}$ be the face of the Birkhoff polytope specified by $b_{12} = b_{21} = 0$, and put $m = n - 2$. Minc's averaging theorem permits a permanent minimizer on this face to be chosen in the form

$$Y = \begin{pmatrix} x_1 & 0 & a_1 \mathbf{1}^\top \\ 0 & x_2 & a_2 \mathbf{1}^\top \\ a_1 \mathbf{1} & a_2 \mathbf{1} & x J_m \end{pmatrix}, \quad x_i = 1 - m a_i, \quad a_1 + a_2 + m x = 1. \quad (5)$$

Here $0 \leq a_i \leq 1/m$. We include the equalization step, since only the averaging theorem is imported.

Proposition 4.1 (One-variable reduction). *For $m \geq 4$, the minimum permanent in (5) occurs with $a_1 = a_2$. Consequently the minimum of $\text{per } B$ on $\Omega_n^{(2)}$ is the minimum, on*

$$\frac{m-2}{m^2} \leq x \leq \frac{1}{m},$$

of

$$P_n(x) = m! x^{m-2} (x^2 a^2 + 2m x a b^2 + m(m-1) b^4), \quad (6)$$

where

$$a = \frac{2 - m + m^2 x}{2}, \quad b = \frac{1 - m x}{2}. \quad (7)$$

Proof. Put $s = a_1 + a_2$, $p = a_1 a_2$, and $x = (1 - s)/m$. Counting permanent terms in (5) gives, apart from the positive factor $m! x^{m-2}$,

$$E_m(s, p) = x^2(1 - m s + m^2 p) + m x(s^2 - (2 + m s)p) + m(m-1)p^2. \quad (8)$$

For fixed s ,

$$\frac{\partial E_m}{\partial p} = 2m(m-1)p + (m+1)s^2 - ms - 1. \quad (9)$$

Since $p \leq s^2/4$ and $0 \leq s \leq 2/m$, the right side is at most

$$D_m(s) = \frac{(m^2 + m + 2)s^2 - 2ms - 2}{2}.$$

This quadratic is convex, while

$$D_m(0) = -1, \quad D_m(2/m) = -1 + \frac{2}{m} + \frac{4}{m^2} < 0 \quad (m \geq 4).$$

Thus (9) is strictly negative. For fixed s , the permanent is minimized by maximizing p , namely by $p = s^2/4$ and $a_1 = a_2 = b$. Solving the stochastic constraints yields (7) and the interval for x . A second count gives (6). $\square$

For $n = 10$, set

$$L_{10} = \frac{36719}{10^8}. \quad (10)$$

Proposition 4.2 (Exact two-zero bound). *Every $B \in \Omega_{10}^{(2)}$ satisfies*

$$\text{per } B > L_{10} > \gamma_{10}.$$

Proof. Here $m = 8$ and the complete interval is $[3/32, 1/8]$. The primary verifier expands $P_{10}(x) - L_{10}$ over $\mathbb{Q}$, verifies positive endpoint values, and uses an exact rational Sturm sequence to prove that it has no root in the interval.

The independent verifier does not use that expansion. It evaluates the permanent of the matrix (7) by the literal exact Ryser formula at eleven rational points. Degree at most ten makes these values determine the polynomial. On each of 256 equal rational subintervals it converts the polynomial to the degree-ten Bernstein basis; every one of the resulting coefficients is strictly positive. This independently proves $P_{10}(x) - L_{10} > 0$ on the entire interval. Finally,

$$L_{10} - \gamma_{10} = \frac{431}{10^8} > 0.$$

$\square$

5 Marginal tight cuts

We first record the elementary stationarity equation used in this case. Since a maximizer is fully indecomposable, all row and column sums are positive.

Lemma 5.1 (Support stationarity). *At a global maximizer, for every positive entry a_{ij} ,*

$$\frac{R}{r_i} + \frac{C}{c_j} - \text{per } A(i \mid j) = \Phi(A), \quad (11)$$

where $A(i \mid j)$ is obtained by deleting row i and column j .

Proof. The left side of (11) is the partial derivative $\partial\Phi/\partial a_{ij}$. At a maximum on the nonnegative simplex, moving a small amount of mass between two positive entries shows that all partial derivatives on the support have a common value μ ; moving mass from a positive entry to a zero entry shows only the corresponding one-sided inequality at a zero. Euler's identity applies because Φ is homogeneous of degree n :

$$n\Phi(A) = \sum_{i,j} a_{ij} \frac{\partial\Phi}{\partial a_{ij}} = \mu \sum_{i,j} a_{ij} = n\mu.$$

Thus $\mu = \Phi(A)$. □

Suppose the tight cut is marginal. The ratio in (3) is then an average of a nonempty set of row sums or column sums. Singleton marginal cuts are also allowed, so

$$\lambda = \min_i r_i \quad \text{or} \quad \lambda = \min_j c_j.$$

By transposition, it is enough to treat a minimum row. Choose i with

$$r_i = \lambda = 1 - a, \quad 0 \leq a < 1.$$

Multiply (11) by a_{ij} and sum over j . Expansion of the permanent along row i gives

$$R + C \sum_j \frac{a_{ij}}{c_j} - P = r_i \Phi(A).$$

Therefore

$$q_i := \sum_j \frac{a_{ij}}{c_j} = 1 - (1 + \theta)a, \quad \theta = \frac{R - P}{C}. \quad (12)$$

By (2) and $P = \gamma_n - \delta$,

$$\theta \geq R - P \geq 1 - \gamma_n. \quad (13)$$

Let $c_{\max} = \max_j c_j$. Since $q_i > 0$, equations (12)–(13) imply

$$\frac{1 - a}{c_{\max}} \leq q_i \leq 1 - (2 - \gamma_n)a.$$

The last denominator is therefore positive, and

$$c_{\max} \geq 1 + h(a), \quad h(a) = \frac{(1 - \gamma_n)a}{1 - (2 - \gamma_n)a}. \quad (14)$$

AM–GM on the other nine row sums, followed by AM–GM on the other nine column sums, now gives

$$R \leq f(a) := (1 - a) \left(1 + \frac{a}{9}\right)^9, \quad (15)$$

$$C \leq g(h(a)) := (1 + h(a)) \left(1 - \frac{h(a)}{9}\right)^9. \quad (16)$$

For the second inequality, the product at fixed largest column sum is bounded by equalizing the other columns; the resulting expression decreases once the largest sum is at least 1.

Define

$$D(a) = 2 - f(a) - g(h(a)). \quad (17)$$

The shared deficit (1) and (15)–(16) imply

$$\delta \geq D(a). \quad (18)$$

Moreover, f is strictly decreasing for $a > 0$, h is strictly increasing, and g is strictly decreasing for positive arguments. Hence D is strictly increasing for $a > 0$. The exact check

$$D(1/50) - \gamma_{10} > 0 \quad (19)$$

and $\delta \leq \gamma_{10}$ confine every feasible a to $0 \leq a < 1/50$.

The dominated doubly stochastic matrix B in (4) inherits the two independent zeroes of A . Proposition 4.2 therefore gives

$$P \geq L_{10}(1-a)^{10}. \quad (20)$$

On the other hand, (18) gives $P = \gamma_{10} - \delta \leq \gamma_{10} - D(a)$. Thus a contradiction follows once

$$H(a) := L_{10}(1-a)^{10} - \gamma_{10} + D(a) > 0 \quad (0 \leq a \leq 1/50) \quad (21)$$

is established.

Put $d(a) = 1 - (2 - \gamma_{10})a$. Clearing the positive denominator $9^9 d(a)^{10}$ in (21) gives a polynomial of degree 20 over $\mathbb{Q}$. The primary verifier checks positive endpoint values and zero roots on the complete interval by an exact rational Sturm sequence. The independent verifier reconstructs the numerator from 21 exact rational values and converts it on $[0, 1/50]$ to the Bernstein basis of degree 26; all 27 coefficients are strictly positive. Both calculations also check (19). In particular,

$$H(0) = L_{10} - \gamma_{10} = \frac{431}{10^8} > 0.$$

This excludes every marginal tight cut.

6 Proper tight cuts

Suppose now that $u, v \geq 1$. Define

$$\varepsilon_r = \sum_{i \in I^c} r_i - u, \quad \varepsilon_c = \sum_{j \in J^c} c_j - v.$$

For positive numbers $x_1, \dots, x_n$ of sum n and product X , binary Pinsker applied after grouping a subset and its complement gives

$$\left(\sum_{i \in S} x_i - |S| \right)^2 \leq -\frac{n}{2} \log X. \quad (22)$$

Applying (22) to the row and column sums, then using Cauchy–Schwarz and (2), yields

$$|\varepsilon_r| + |\varepsilon_c| \leq \sqrt{-n \log(RC)} \leq \sqrt{\frac{n\delta}{1-\delta}} =: T. \quad (23)$$

The cut identity

$$s(A[I, J]) = k - \varepsilon_r - \varepsilon_c + s(A[I^c, J^c])$$

therefore implies

$$\lambda \geq 1 - \frac{T}{k}. \quad (24)$$

For $n = 10$, the exact inequality $n\gamma_n < 1 - \gamma_n$ ensures $T < 1 \leq k$.

Because B is doubly stochastic,

$$s(B[I, J]) = k + s(B[I^c, J^c]) \geq k.$$

Tightness and (4) give the reverse bound after multiplication by λ , so equality holds and

$$B[I^c, J^c] = 0. \quad (25)$$

Thus B lies on the staircase face with a prescribed $u \times v$ zero block. Hwang's theorem says that the barycenter minimizes the permanent on this face. Its permanent is

$$\text{per } B \geq \nu_{n;u,v} := \frac{\gamma_{n-u}\gamma_{n-v}}{\gamma_k}, \quad k = n - u - v, \quad (26)$$

where $\gamma_0 = 1$. Indeed, the barycenter has $(n-v)!(n-u)!/k!$ supported permutations, all of the same weight; both verifiers reconstruct this count exactly.

Using (24), Bernoulli's inequality, and $1 - \delta \geq 1 - \gamma_n$ gives

$$\nu_{n;u,v} \left(1 - \frac{T}{k}\right)^n - (\gamma_n - \delta) \geq \nu_{n;u,v} - \gamma_n + \delta - \frac{n\nu_{n;u,v}}{k} \sqrt{\frac{n\delta}{1 - \gamma_n}}.$$

Completing the square in $\sqrt{\delta}$ gives the uniform lower bound

$$\eta_{n;u,v} := \nu_{n;u,v} - \gamma_n - \frac{n^3 \nu_{n;u,v}^2}{4k^2(1 - \gamma_n)}. \quad (27)$$

Both verifiers check $\eta_{10;u,v} > 0$ for all

$$u, v \geq 1, \quad u + v \leq 9,$$

namely all 36 proper cuts. The smallest value occurs at $(u, v, k) = (1, 1, 8)$:

$$\nu_{10;1,1} = \frac{75161927680}{205891132094649},$$

and

$$\eta_{10;1,1} = \frac{171312851843447795657879293024418789}{103456482841067610005769448332714895312500} > 0.$$

This contradicts $P = \gamma_n - \delta$ in every proper-cut case.

The whole, marginal, and proper cases exhaust every tight cut. No boundary global maximizer exists. Hwang's positive-support theorem identifies the sole remaining maximizer as $J_{10}/10$, proving Theorem 1.1.

7 Exact computational audit

No floating-point value is used in a correctness decision. The primary standard-library verifier checks:

1. the bivariate equalization identity (9);
2. the permanent polynomial (6);
3. the exact lower bound (10) by a rational Sturm sequence;

4. the degree-20 numerator of (21), denominator positivity, endpoint positivity, and its exact zero root count;
5. the staircase barycenter count and all 36 values in (27).

The independent verifier does not import the primary verifier and shares neither of its positivity procedures. It uses a literal exact Ryser sum and Lagrange interpolation for the two-zero polynomial, followed by Bernstein positivity on 256 rational subintervals. It reconstructs the marginal numerator by exact value interpolation and proves it positive using a degree-26 Bernstein expansion. Its smallest certified Bernstein coefficients are, respectively,

$$\frac{2098318974119444468161384333}{3961408125713216879677197516800000000} > 0$$

and

$$\frac{107851173950140635743821359}{39672851562500000000000000} > 0.$$

It separately checks the equalization identity, $D(1/50) > \gamma_{10}$, the staircase counts, and all proper-cut inequalities.

The integration test runs both verifiers in ordinary and optimized Python mode and requires identical output, confirming that no correctness decision depends on an executable `assert` statement. It also replaces L_{10} by a false inflated bound and requires both implementations to reject the mutation.

8 Scope and status

This note establishes dimension 10. Together with the other exact working proofs in the accompanying repository, the currently covered dimensions are

$$4, 5, 10, 11, 12, 13, 14, 15, 16.$$

Dimensions 6, 7, 8, 9 are not settled by these materials. Pang’s June 2026 preprint treats $n \geq 17$ [8]. The present result should be described as an exact computer-assisted working proof until the mathematical reduction and both programs have undergone independent peer review.

Reproducibility

From the `n10` directory, run

```
python3 -I verify_dittert_n10.py
python3 -I audit_dittert_n10_stdlib.py
python3 -I test_n10_package.py
```

The verifier final lines must be

```
CERTIFIED
INDEPENDENT AUDIT CERTIFIED
```

and the tests must report OK.

References

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